NATIONAL SCHOOL OF BUSINESS MANAGEMENT

Model Paper

SE204.3- Development of Enterprises Applications

Instructions to Candidates

1) **Answer All Questions (Part A & B)**
2) **In part C, out of 4 questions, answer ONLY 3 questions**
3) Illustrate answers with sketches where required.
4) The time allocated for the examination is **three (03) hours.**
5) If a page or a part of this question paper is not printed, please inform the Supervisor immediately.
6) Write your index number on all pages of the answer script.

Part A

[1*30 = 30 marks]

1. A desktop payroll application performs UI rendering, salary calculation, and file storage within the same system without server communication. Which architectural style best describes this?
   a. Two-tier
   b. Three-tier
   c. One-tier
   d. Microservices
   e. Distributed

2. A web system separates presentation in the browser, business logic on an application server, and data storage on a database server. This most accurately represents:
   a. Monolithic architecture
   b. Three-tier architecture
   c. One-tier architecture
   d. Embedded system
   e. File-based system

3. A company deploys its entire application as a single executable package. Updating one module requires redeploying the entire system. This indicates:
   a. Microservices
   b. Layered distributed architecture
   c. Monolithic architecture
   d. Service mesh
   e. Stateless architecture

4. An application must scale only the payment service independently during high transaction periods. Which architecture best supports this requirement?
   a. Monolithic
   b. One-tier
   c. Microservices
   d. Embedded
   e. Static
5. An endpoint is designed as: `DELETE /api/v1/products/15` The operation most likely:
   a. Retrieves product 15
   b. Updates product 15
   c. Deletes product 15
   d. Creates product 15
   e. Disables all products

6. A developer defines endpoints such as:
   ```
   /createOrder
   /getOrder
   /deleteOrder
   ```

Which REST design principle is most likely violated?
a. Statelessness
b. Resource-based URL design
c. Layered system
d. Client-server separation
e. Caching

7. A system requires each request to contain authentication data, and the server does not store login state. This system is:
   a. Stateful
   b. Session-based
   c. Cookie-dependent
   d. Stateless
   e. Server-bound

8. If a server restart causes all logged-in users to lose access, the system most likely uses:
   a. JWT authentication
   b. Stateless tokens
   c. Session-based authentication
   d. OAuth2
   e. API keys

9. In Spring Boot, which component should primarily handle HTTP request mapping?
   a. Service
   b. Repository
   c. Controller
   d. Entity
   e. DTO

10. Which annotation indicates that a class represents a database table?
    a. @Service
    b. @RestController
    c. @Entity
    d. @Component
    e. @Transactional

11. A class responsible for business rules should be annotated with:
    a. @Repository
    b. @Service
    c. @RestController
    d. @Entity
    e. @Id

12. A Spring Boot application successfully connects to the database locally. However, when deployed to production, it fails to connect because the database credentials are different.

Which file is most appropriate for externalizing and managing such environment-specific configuration?

a. pom.xml
b. application.properties
c. SecurityFilterChain.java
d. Entity class
e. Controller class

13. Which layer should directly communicate with the database?
    a. Controller
    b. Service
    c. DTO
    d. Repository
    e. View

14. Which of the following best represents correct request flow?
    a. Controller → Repository → Service → DB
    b. Frontend → Service → Controller → DB
    c. Frontend → Controller → Service → Repository → DB
    d. DB → Controller → Service
    e. Service → Frontend → Repository

15. JWT consists of which three components?
    a. Header, Body, Cookie
    b. Header, Payload, Signature
    c. Payload, Token, Hash
    d. Cookie, Claim, Signature
    e. Header, Token, Session

16. Which component ensures integrity of JWT?
    a. Header
    b. Payload
    c. Signature
    d. Claim
    e. Cookie

17. Which storage mechanism persists even after browser restart?
    a. Session
    b. SessionStorage
    c. LocalStorage
    d. Server Memory
    e. Temporary Cache

18. Cookies are limited approximately to:
    a. 512 bytes
    b. 1 KB

c. 4 KB
d. 10 KB
e. Unlimited

19. Which annotation enables cross-origin access?
    a. @Service
    b. @Autowired
    c. @CrossOrigin
    d. @Query
    e. @Transactional

20. In layered architecture, which responsibility should NOT belong to Controller?
    a. Receiving HTTP request
    b. Returning HTTP response
    c. Applying business rules
    d. Calling service methods
    e. Mapping endpoints

21. A developer designs an API where the client must call /login before accessing any other endpoint. The server stores user state in memory. After deployment across multiple servers, users randomly get logged out. What is the most probable cause?

    a. Incorrect URL mapping
    b. Lack of HTTPS
    c. Server-side state dependency
    d. Incorrect HTTP method usage
    e. Missing DTO layer

22. An endpoint is defined as: `PUT /api/v1/customers/20` The request body contains only the customer's email address.Which issue is most likely to occur if designed incorrectly?
    a. Complete replacement of the resource
    b. Violation of statelessness
    c. Token invalidation
    d. Session expiration
    e. CORS conflict

23. A Spring Boot application contains the following issue: Business validation logic is written directly inside a controller method. What architectural principle is being compromised?
    a. Statelessness
    b. Resource-based design
    c. Layer separation
    d. Client-server model
    e. JWT integrity

24. A team wants to customize SQL queries for complex filtering logic inside a repository interface. Which annotation is most appropriate?

a. @RestController
b. @Transactional
c. @Query
d. @Autowired
e. @Component

25. An enterprise API must support multiple client types (web, mobile, third-party integrations) without modifying backend authentication logic. Which authentication approach best supports this?
    a. Server sessions
    b. In-memory authentication
    c. JWT-based authentication
    d. LocalStorage-only authentication
    e. Cookie-only authentication

26. A Spring Boot application is configured to automatically update the database structure when the application starts. What is the most likely result?
    a. The database will be deleted
    b. The database structure may change automatically
    c. The application will stop running
    d. The server port will change
    e. The API endpoints will be removed

27. A developer packages the application as a WAR file instead of a JAR. This implies:
    a. The application cannot expose REST APIs
    b. It must run without any server
    c. It requires an external application server
    d. It cannot connect to databases
    e. It becomes stateless automatically

28. An application stores JWT tokens in browser localStorage. Later, it becomes vulnerable to XSS attacks. Why is this risky?
    a. Tokens are too large
    b. localStorage is encrypted
    c. JavaScript can access stored tokens
    d. Tokens expire immediately
    e. HTTP methods change automatically

29. A Spring Boot service method performs multiple database updates that must either all succeed or all fail together. Which annotation ensures this behavior?
    a. @Query
    b. @CrossOrigin
    c. @Transactional
    d. @RestController
    e. @Entity

30. An API design exposes internal database table names directly in URL paths such as: `/tbl_user_data_all_records` Which issue does this most likely indicate?
    a. Strong encapsulation
    b. Poor abstraction and tight coupling
    c. Stateless design
    d. Proper REST compliance
    e. Efficient caching

Part B

Answer All the questions. [Each questions carries 4 marks]

1. Explain the difference between One-Tier, Two-Tier and Three-Tier architectures. Give one practical example for each.
2. Explain the responsibilities of Model, View, and Controller in MVC architecture.
3. Identify two limitations of Monolithic architecture and explain how Microservices address them.
4. Explain what makes an API RESTful. Discuss the importance of using proper HTTP methods in API design.
5. Describe the difference between GET, POST, PUT, and DELETE methods. Provide one practical use case for each.
6. Explain the layered architecture used in Spring Boot applications. Describe the responsibility of Controller, Service, and Repository layers.
7. Explain the purpose of the following Spring Boot annotations:
    a. @RestController
    b. @Service
    c. @Repository
    d. @Entity
8. Describe how a Spring Boot application connects to a database. Mention key configuration properties required.
9. Explain the difference between Sessions and Cookies. Why is HTTP considered stateless?
10. Explain how JWT authentication works in a Spring Boot application. Describe the structure of a JWT token

Part C

Answer ONLY 3 questions [Each question carries 10 marks]

1. A company currently runs a monolithic e-commerce application. As user traffic increases, the system becomes slow, difficult to deploy, and hard to maintain. Different teams are working on payments, inventory, and user management, but deployments affect the entire system. You are asked to propose a better architectural solution. Discuss:

- Limitations of the current architecture
- How a microservices architecture would improve scalability and deployment
- Possible challenges during migration
- When monolithic architecture may still be appropriate

2. A development team creates the following endpoints:
   - `/getAllUsers`
   - `/deleteUserById`
   - `/updateUserDetails`

   These endpoints work correctly but the system must now be exposed to external clients. Critically evaluate this API design. Discuss:

   - REST principles that may be violated
   - Importance of resource-based URLs
   - Proper use of HTTP methods
   - How the API should be redesigned

3. A Spring Boot application contains the following issues: Business logic written inside controllers, Direct database access inside controllers, No clear separation between layers. The system works but becomes difficult to maintain and test. Discuss:

   - Why this design is problematic
   - How proper layered architecture improves maintainability
   - Responsibilities of Controller, Service, and Repository layers
   - Long-term benefits of refactoring

4. An enterprise web application currently uses session-based authentication. The system must now:
   - Support mobile clients
   - Scale across multiple servers
   - Avoid losing login state during server restarts

   You are asked to redesign the authentication mechanism. Discuss:

   - Limitations of session-based authentication
   - How JWT-based authentication works
   - Advantages of stateless authentication
   - Security risks and best practices when using JWT

Write the answer number for Questions 1 to 30 (a, b, c, d or e) in the table below. **[30 Marks]**

1		**16**	
2		**17**	
3		**18**	
4		**19**	
5		**20**	
6		**21**	
7		**22**	
8		**23**	
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12		**27**	
13		**28**	
14		**29**	
15		**30**	